Deep Learning Course Final Assignment

Final Project: Applying Neural Networks to a Real Dataset

For the final assignment, students should work in groups of **2–3 members**. Each group should choose a domain of interest, select a real dataset from that domain, and apply neural network models learned during the course.

The dataset should be meaningful and non-trivial. It should not be a very small toy dataset. Students may use datasets from Kaggle, UCI Machine Learning Repository, Hugging Face Datasets, TensorFlow Datasets, or other reliable public sources.

Project Registration

Each group should fill in the following information in the Google Sheet:

- Group member names
- Project topic
- Dataset link

Google Sheet link:

Project Requirements

Each group should:

1. Choose a real-world domain and dataset.
2. Clearly define the problem they want to solve.
3. Prepare and preprocess the dataset.
4. Train one or more neural network models.
5. Evaluate the trained models using suitable metrics.
6. Compare the results and discuss the findings.
7. Present the final results to the class.

Students may use any neural network architecture covered in the course, including:

- Fully Connected Neural Networks for tabular data
- Convolutional Neural Networks for image data
- CNNs with transfer learning for image data
- Recurrent Neural Networks, LSTM, or GRU for sequence, text, or time series data

Final Presentation

Each group should prepare a short presentation (10-15 min) that includes:

- Project title and group members
- Chosen domain and dataset
- Problem definition
- Data preprocessing steps
- Neural network model architecture
- Training and evaluation results
- Comparison of experiments
- Discussion, conclusions, and possible improvements

Deadline

The expected deadline is **around June 25**. The exact deadline will be finalized and adjusted later.

Important Notes

The goal of the assignment is to demonstrate understanding of the deep learning workflow: choosing a suitable dataset, preparing the data, training neural network models, evaluating results, and discussing the strengths and limitations of the approach.